FINAL DRAFT

Solar Filament Segmentation Challenge 2026

Verified 0.40 public leaderboard pipeline

Kaggle status	**COMPLETE - public score 0.40**
Submission	1,261 rows across 180 blind test images
Canonical SHA-256	4ee79cb588f9415eb2f48ebb1a129382f0d7f5e6410724be8b9f6b213b18e512
Private GitHub	github.com/kulharshit21/solar-filament-final-draft
Drive artifacts	Weights Competion Submmsion

This report documents the full work: evaluator reconstruction, integrity controls, date-grouped validation, model architecture, training strategy, OOF calibration, test inference, submission validation, Drive/GitHub separation, and the final Kaggle result.

Integrity note: the historical inference generator was audited locally. Its original blind-test caches and data/test_images directory are no longer present in the recovered archive, so it stopped safely with a missing-input error. The exact historical CSV was preserved, verified byte-for-byte, and independently resubmitted to Kaggle with a returned score of 0.40. No new CSV was fabricated.

1. Executive summary

The pipeline segments elongated, low-contrast solar filaments in H-alpha images. The official evaluator compares each prediction against multiple human raters independently, making the precision/recall balance and structural quality of each instance critical. The final verified package uses the recovered model weights and the calibrated post-processing policy that produced the 0.40 leaderboard result.

```
official training data
|
v
date-grouped folds + annotator-aware targets
|
+--> YOLO11x-seg: high-resolution local proposals
+--> Mask2Former Swin-Large: global topology/context
+--> SAM 2.1 Hiera-L: box-prompted boundary refinement
|
v
OOF calibration -> fusion -> disjoint masks -> official RLE CSV
```

2. Work chronology

Audit and recovery

1. Inspected the organizer evaluation logic and reproduced local metric behavior.

2. Recovered the historical clean baseline and compared prior submissions, weights, reports, and manifests.

3. Established a strict no-leak rule: no test labels, hidden annotations, cluster metadata, or prohibited artifacts.

4. Backed up project artifacts to Drive and removed only project-specific temporary DGX material after checksums passed.

Validation design

1. Parsed the official annotation JSON.

2. Grouped observations by date to avoid temporal leakage across fold boundaries.

3. Kept rater views explicit rather than collapsing disagreement into a synthetic label.

4. Measured TP, FP, FN, PQ, SQ, RQ, sum-IoU, overlap pixels, and RLE decodability.

5. Selected a stable OOF plateau instead of a one-off validation spike.

Training and engineering

1. Prepared high-resolution train/test inputs and fold manifests.

2. Trained and evaluated YOLO11x-seg for local high-resolution proposals.

3. Fine-tuned Mask2Former Swin-Large for global structure.

4. Fine-tuned the SAM 2.1 decoder for box-prompted boundary refinement.

5. Ran multi-GPU inference in strict waves when only physical GPUs 5 and 6 were allowed.

6. Built OOF and test candidate banks, calibrated the selector, and validated the final submission.

7. Uploaded artifacts with incremental and final checksum synchronization.

3. Architecture

Data and target layer

• Official training images and annotations only.

• Each rater view remains a separate supervision instance.

• Date-grouped folds prevent temporal leakage.

• Minimum-area statistics are computed from training annotations only.

YOLO11x-seg detection layer

A high-resolution YOLO11x segmentation model supplies dense local filament instances, confidence scores, and precise small-structure proposals. This path is important for narrow spines and barbs and for suppressing irrelevant chromospheric texture.

Mask2Former global segmentation layer

Mask2Former uses a COCO-initialized Swin-Large pixel backbone and transformer mask decoder. It sees the full resized image and contributes long-range topology and context when local contrast is ambiguous. The historical canvas was 1280 pixels.

SAM 2.1 boundary layer

SAM 2.1 Hiera-L receives candidate boxes and refines masks on context crops. The historical design fine-tunes the decoder while keeping the expensive image encoder frozen, preserving narrow absorption-valley boundaries without aggressive erosion.

Fusion and selection layer

YOLO-family candidates are averaged, then combined with native Mask2Former candidates at equal weight. OOF calibration selects the confidence, mask, and area thresholds. Overlap arbitration guarantees that output instances are pixel-disjoint before official RLE encoding.

4. Locked 0.40 policy

Parameter	Value
Variant	ens[avg[yolo]|mask2former;w=1:1]
Mask threshold	0.45
Score threshold	0.53
Minimum area	319 pixels (training p05)
Maximum area	70,502 pixels
CRF	Disabled
Overlap rule	Disjoint pixel arbitration

5. Evidence and final metrics

Metric	Value
OOF PQ	0.4622
OOF SQ	0.6857
OOF RQ	0.6741
OOF TP / FP / FN	1031 / 454 / 543

Test images	180 / 180
Submission instances	1,261
Overlap pixels	0
Undecodable RLEs	0
Kaggle public score	0.40, COMPLETE

6. Reproduction and release procedure

1. Install the requirements in the private repository.

2. Place only official competition data under the configured data root.

3. Run the code-only integrity verifier and official preflight.

4. Download the separate Drive artifacts and verify SHA256SUMS.txt.

5. Run preparation, training, prediction, and OOF calibration scripts.

6. Freeze the locked selector policy and build the test CSV.

7. Validate schema, unique IDs, RLEs, image coverage, and disjoint pixels.

8. Record hashes and submit only the validated CSV.

```
python scripts/verify_code_release.py
python -m compileall -q scripts src
python scripts/00_verify.py
python scripts/01_prepare.py
python scripts/05_predict.py --split test --source yolo --overwrite
python scripts/05_predict.py --split test --source mask2former --overwrite
python scripts/05_predict.py --split test --source sam2 --overwrite
python scripts/generate_w11.py
```

7. Integrity boundary

• No test labels were used for training or calibration.

• No hidden organizer annotations or prohibited cluster files are included.

• GitHub contains code and documentation only.

• Drive contains weights, exact submission, evidence, and manifests.

• Credentials and Kaggle API tokens are excluded.

• The canonical CSV is hash-locked and the Drive copy has zero checksum differences.

• The 0.38 Python4 candidate is excluded from the final draft.

8. Final handoff links

Private GitHub code: https://github.com/kulharshit21/solar-filament-final-draft

Drive weights and submission: https://drive.google.com/open?id=1qikTNZkhUWEC3vB0Yz2_5l5Zwr1AehFN

Competition: https://www.kaggle.com/competitions/filament-segmentation-2026

Artifact verification result: 9 canonical files checked, 1,261 submission rows, 180 test images, and the exact 0.40 CSV hash confirmed.